

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Source: arxiv.org/abs/2504.03553

Autonomously regulating when to act, reflect, or retrieve knowledge

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Core issue: Existing methods inject trajectories/feedback/knowledge regardless of situational need, unlike human self-aware regulation.

Trajectory Imitation

- SFT on gold traces
- Assumes expert trajectories always needed
- Costly, low adaptability

Trial-and-Error Reflection

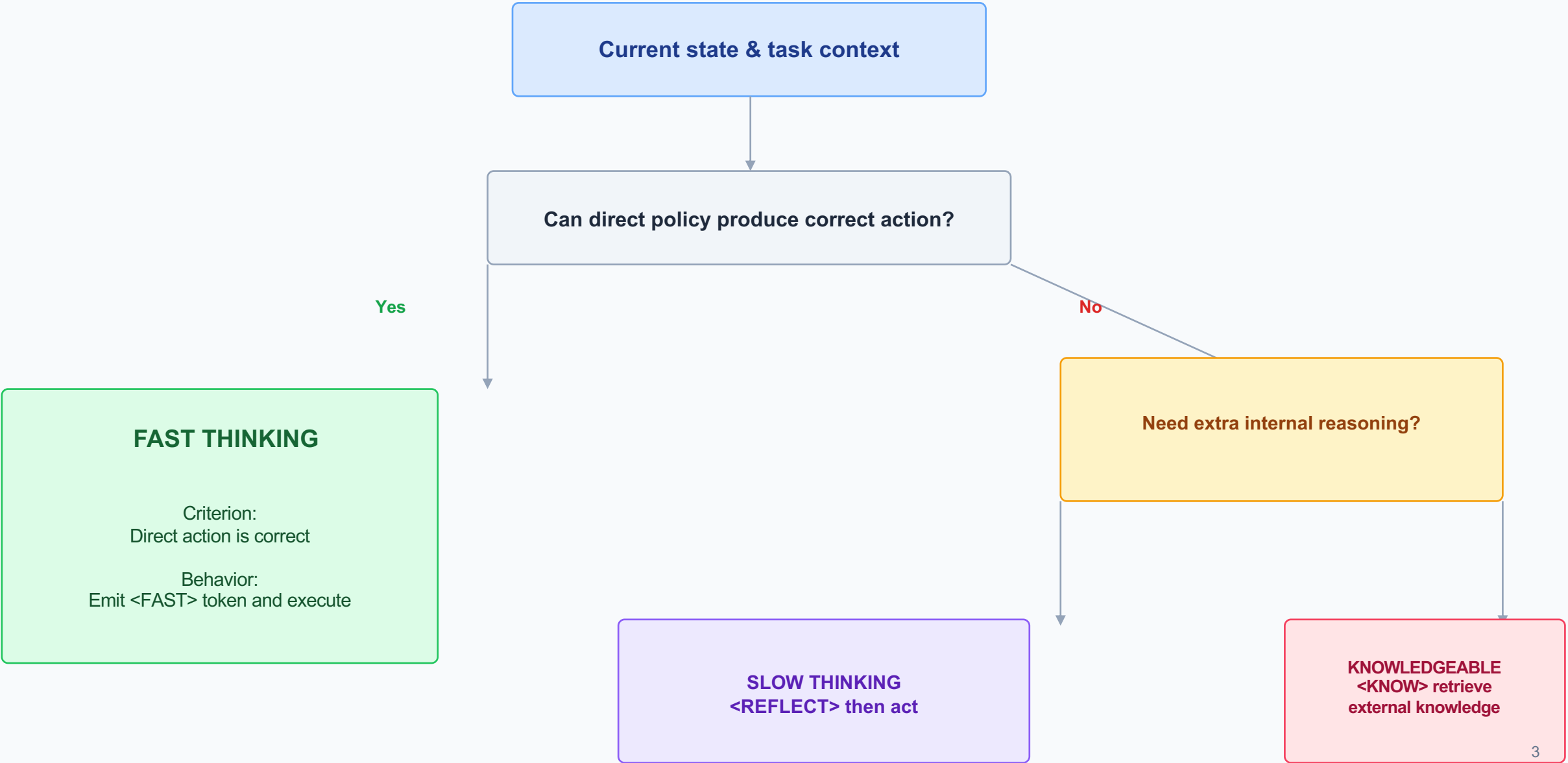
- Reflect from failed attempts
- Blindly re-plans every case
- Extra tokens on easy tasks

Knowledge-Augmented Planning

- Inject external knowledge at all times
- 100% knowledge querying in many setups
- Noise + retrieval overhead

Need: Situation-aware
cognitive control

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Framework reference: figure_3.jpg

1) Data Construction

- Self-explore ALFWorld trajectories
- Heuristic situation judgment labels each step
- Insert special tokens: <FAST>, <REFLECT>, <KNOW>
- No expensive human step-level annotation

2) Two-Stage Training

- Stage A: Supervised fine-tuning (SFT)
- Learn action + situation token alignment
- Stage B: RPO optimization
- Reinforce adaptive decision quality

3) Self-Aware Inference

- Model predicts situation token first
- <FAST>: execute directly
- <REFLECT>: trigger internal reflection
- <KNOW>: retrieve then continue planning

Key mechanism: special tokens operationalize metacognitive control over reasoning depth and knowledge retrieval.

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Qualitative reference: figure_1.jpg

Task A: Put a hot mug in cabinet

KnowSelf behavior:

- Predicts reflection-needed situation
- Performs heating sub-steps before placement
- Completes full goal sequence successfully

OpenAI-O1 baseline:

- Misses heating prerequisite
- Fails task completion

Interpretation:

Self-reflection mode helps recover implicit procedural requirements that direct action misses.

Task B: Put a saltshaker in drawer

KnowSelf behavior:

- Detects external-knowledge-needed situation
- Retrieves relevant placement knowledge
- Executes correct placement action

DeepSeek-R1 baseline:

- Gets stuck in ineffective loops
- Fails to locate correct strategy

Interpretation:

Knowledgeable-thinking mode prevents blind trial-and-error when missing world knowledge.

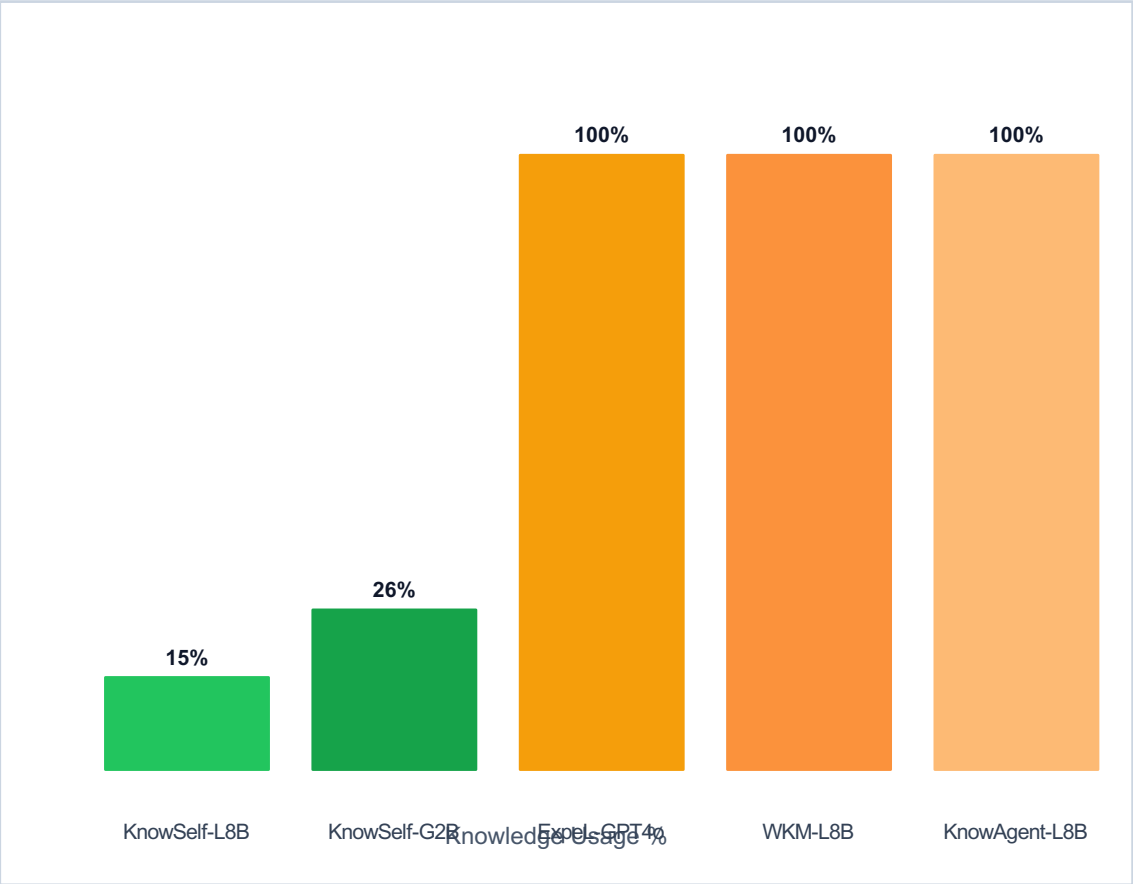
KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Method	Backbone	Know%	Clean	Cool	Heat	Examine	Pick	Put	All
ReAct	GPT-4o	-	67.12	58.82	64.29	66.67	82.35	81.06	69.40
Reflexion	GPT-4o	-	71.23	58.82	64.29	71.43	76.47	75.76	69.40
ExpeL	GPT-4o	100	69.86	67.65	66.67	76.19	88.24	84.85	79.85
ETO	Llama-8B	-	63.01	58.82	69.05	64.29	58.82	77.27	67.16
KnowAgent	Llama-8B	100	63.01	58.82	66.67	57.14	52.94	72.73	63.43
WKM	Llama-8B	100	69.86	70.59	71.43	59.52	76.47	81.82	73.13
KnowSelf	Llama-8B	15	79.45	79.41	85.71	85.71	82.35	93.94	84.33
ETO	Gemma-2B	-	65.75	52.94	64.29	69.05	64.71	75.76	66.42
KnowAgent	Gemma-2B	100	69.86	64.71	66.67	66.67	52.94	74.24	68.66
WKM	Gemma-2B	100	64.38	70.59	69.05	73.81	58.82	80.30	70.90
KnowSelf	Gemma-2B	26	71.23	67.65	73.91	76.19	83.33	52.94	79.85

KnowSelf (Llama-8B): best overall accuracy (84.33) with only 15% knowledge usage; full-knowledge baselines use 100%.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods

Knowledge usage (%) vs overall ALFWorld accuracy (All)



Accuracy with selective knowledge		
KnowSelf-Llama-8B	All: 84.33	Know 15%
KnowSelf-Gemma-2B	All: 79.85	Know 26%
ExpeL-GPT-4o	All: 79.85	Know 100%
WKM-Llama-8B	All: 73.13	Know 100%
KnowAgent-Llama-8B	All: 63.43	Know 100%

Takeaway: 15–26% knowledge access is sufficient for top performance; 100% querying is not only costly but often less accurate.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful and can reduce planning performance.

2

KnowSelf learns situational self-awareness via lightweight heuristic token labeling on self-explored trajectories.

3

KnowSelf (Llama-8B) surpasses strong baselines: 84.33% vs GPT-4o ReAct 76.12% and GPT-4o ExpeL 79.85%.

4

Only 15–26% knowledge usage is needed, compared with 100% in full-knowledge methods.

5

The paradigm scales across backbones: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions: extend situational self-awareness to open-world tools, multi-agent coordination, and continual adaptation under distribution shift.